12/12/24 M/S

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THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2024

COMPUTATIONAL STATISTICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	List the range difference of the Correlation Coefficient Formula and the covariance Formula?	L1	CO1	**02**
	b.	Define Eigen-decomposition.	L1	CO3	**02**
	c.	State the following **i.** similarity metric **ii.** distance metric.	L1	CO3	**02**
	d.	List types of hierarchical clustering.	L1	CO5	**02**
	e.	Define unsupervised learning.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain multivariate normal distribution, and its key properties?	L2	CO1	**05**
		OR			
	b.	List out the difference between univariate, bivariate, and multivariate analysis.	L1	CO1	**05**
3.	**a.**	Define Principal Component Analysis (PCA)? After finding the principal components as eigenvectors, how are the input matrix's dimensions reduced?	L2	CO2	**05**
		OR			
	b.	Explain how the covariance matrix is generated. What is the significance of the covariance matrix?	L2	CO2	**05**
4.	**a.**	Describe how the linear discriminate analysis can be more effective for pre-processing discrete labeled data.	L2	CO3	**05**
		OR			
	b.	Explain what role eigen-decomposition plays in dimensionality reduction.	L2	CO3	**05**
5.	**a.**	Discuss factor analysis with example.	L2	CO4	**5**
		OR			
	b.	List out all the main types of Factor Analysis?	L1	CO4	**05**
6.	**a.**	Explain K-Means Clustering and how does it work?	L2	CO5	**05**
		OR			
	b.	State the steps for divisive hierarchical clustering with example?	L1	CO5	**05**

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7.	**a.**	Explain the limitations of K-Means Clustering?	L2	CO5	**05**
		OR			
	b.	Differentiate between Factor Analysis and Principal Component Analysis (PCA).	L4	CO2	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Compare bivariate analysis and multivariate analysis.	L4	CO1	**10**
		OR			
	b.	Explain,it important for features/attributes to follow a normal distribution in data analysis?	L3	CO1	**10**
9.	**a.**	Explain,how do you decide how many principal components to keep in PCA?	L2	CO2	**10**
		OR			
	b.	Explain the main benefits of using PCA?	L2	CO2	**10**
10.	**a.**	Elaborate on how the scatter matrix is generated for linear discriminate analysis.	L2	CO3	**10**
		OR			
	b.	Outline difference between PCA and LDA.	L1	CO3	**10**
11.	**a.**	Give comparison between covariance calculation and scatter matrix calculation in dimensionality reduction. And also highlight there significances.	L4	CO4	**10**
		OR			
	b.	Can dimensionality reduction be used to increase corelation between features and sample labels. Present a case study.	L3	CO4	**10**
12.	**a.**	State the steps for agglomerative hierarchical clustering with example?	L1	CO5	**10**
		OR			
	b.	What are the main linkage methods used in agglomerative and divisive hierarchical clustering?	L2	CO5	**10**
13.	**a.**	How can you check if a feature follows a normal distribution, and what can you do if it does not?	L3	CO1	**10**
		OR			
	b.	What is center limit theorem. Explain with example.	L2	CO1	**10**